

Supplementary mathematical material

Case 4: the S^rT^s component

Formal proof for bounded-depth competing architectures

Scope. This document supplies the dense two-run component used in the insufficient-profile analysis. For each fixed depth bound D , it constructs a task solved exactly by S^rT^s with logarithmic state and token budgets and window $m + 1$. Every depth-at-most- D profile avoiding S^rT^s has error $1/2 - o(1)$ under a common constant-factor enlargement of these budgets.

The proof separates the architectural reduction from the random-table argument. The latter includes a physical replay definition, uniform finite probability estimates, posterior capture, causal training tests, bottom-up prefix alignment, the complete first-query label, and the optimized crossed continuation. No unrestricted advice tape or decoder of the complete raw future is granted.

Reading guide. Section 1 states the component theorem and the exact conventions. Sections 2–5 give the finite ingredients. Sections 6–8 prove the mutual capture/program induction. Section 9 treats the non-permutation first stage, and Section 10 proves the component theorem. The appendices record the jointly causal latent-state interface and the use of this result in the shorter manuscript argument.

Contents

1	Statement, conventions, and proof structure	2
2	Parameters and an explicit auxiliary table family	2
3	Physical replay systems and their closure	4
4	Uniform finite random-table estimates	6
5	Deterministic and probabilistic bookkeeping lemmas	10
6	Chronological normalization of a replay	13
7	Training dictionaries and bottom-up alignment	15
8	The mutual program/capture induction	18
9	The iid first stage and arbitrary source chunks	21
10	The task, the architectural reduction, and completeness	23
A	Jointly causal latent-state simulation	26
B	Interface with the shorter manuscript argument	27

1 Statement, conventions, and proof structure

All logarithms in bit budgets are base two. Logarithms in exponential probability estimates may be natural; this changes only absolute constants. The depth bound D is fixed, and every asymptotic statement is as $m \rightarrow \infty$, after all constants depending on D have been chosen. Bounds denoted $o(1)$ are uniform over the competing maps in the stated class.

A T-layer of window W outputs one token at position t as an arbitrary function of the preceding layer's tokens at positions $\max\{1, t - W + 1\}, \dots, t$. A S-layer has a finite state set $\mathcal{S}$, an update map, and an emission map depending only on its updated state. Its bit budget is $\log_2 |\mathcal{S}|$. Every intermediate token has at most P bits. We allow position-dependent maps in the lower bound. All maps may be nonuniform and selected after the task tables. This only strengthens the competing class relative to using fixed maps and explicit position tags.

Let $\Sigma = \{S, T\}$. For profiles $A, B \in \Sigma^*$, write $A \preceq B$ for subsequence containment with multiplicities retained. Write $\mathcal{M}_A(B, P, W)$ for the corresponding resource-bounded models. Outputs are evaluated at the final physical position. Fixed left padding and a branch tag in every token are permitted as part of the task definition.

Theorem 1.1 (Dense two-run component). *For every integer $D \geq 2$, there are constants $C_D > 0$ and $\lambda_D > 1$, common to all positive r, s with $r + s \leq D$, such that the following holds for all sufficiently large m . There are a deterministic task $F_{r,s,m}$ and an input distribution $\nu_{r,s,m}$ on $OD(m)$ tokens over an alphabet of size $\text{poly}_D(m)$ with these properties.*

1. A model in $\mathcal{M}_{S^rT^s}(b_m, b_m, m + 1)$ solves the task exactly, where $b_m = \lceil C_D \log_2 m \rceil$.
2. For every $A \in \Sigma^{\leq D}$ with $S^rT^s \not\preceq A$, every model in $\mathcal{M}_A(\lfloor \lambda_D b_m \rfloor, \lfloor \lambda_D b_m \rfloor, \lfloor \lambda_D(m + 1) \rfloor)$ has error at least $1/2 - o(1)$ under $\nu_{r,s,m}$.
3. The same task and distribution work for input-independently randomized models.

The task and all its tables are fixed before the competing profile, maps, and random seed are selected.

In particular, the error in part 2 eventually exceeds $1/3$. The theorem concerns the ambient component, not the separate combination with the wrong-order construction.

The two branches. The task is an equal mixture of a chain branch and a reach branch. A competitor with fewer than s T-layers fails on the reach branch. Every remaining avoiding competitor fails on the chain branch. The unassigned branch is allowed success one; thus it is the *mixture error*, not mixture success, that approaches $1/2$.

Logical dependencies. The proof uses two different assertions. G_j is hardness of a physical j -lookup permutation program with a uniform initial key. $\mathcal{C}_j$ captures the already-arrived prefix of the first unresolved query in a replay system. The induction proves G_j from lower-stage statements, then $\mathcal{C}_j$ using its now-fixed first-query length. The source of the final component has an iid, non-permutation first table; that stage is treated separately by repeating the first-stage certificate construction, not by applying a permutation theorem to an iid table.

2 Parameters and an explicit auxiliary table family

Put $d = \max\{D, 5\}$ and $L_{\text{occ}} = 100d^2$. The symbol d is only an auxiliary envelope: the moving-cut argument for an actual competitor still uses its actual depth bound D .

2.1 Rate and entropy constants

Set $c_1 = 1/16$, $N_1 = 100d$, $A_1 = 2d$, and $s_0 = 100(d+1)$. At level $2 \leq j \leq d$, put $c = \min_{i < j} c_i$, choose the integer $g_j = \lceil 2^{d+10}(d+2)/c \rceil$, and put $b_j^{\text{tr}} = 2g_j$. Here b_j^{tr} is a training-tree arity, not the model bit budget b_m .

Let $R_j^{\text{max}} = (b_j^{\text{tr}})^d$, $p_j = d+1$, $u_j = c/8$, and $c_j = u_j/(400p_j R_j^{\text{max}})$. A level- j capture tree has height j , while a level- j program certificate samples only the remaining future and has height $j-1$. For either relevant height h , its numbers of leaves and internal nodes are $R = (b_j^{\text{tr}})^h$ and $I = (R-1)/(b_j^{\text{tr}}-1)$. Always $R \leq R_j^{\text{max}}$.

The entropy constants are chosen in the order $(N_1, A_1), (N_2, A_2), \dots$. With $N_{<j} = \max_{i < j} N_i$ and $A_{<j} = \max_{i < j} A_i$, define them by the following recursion:

$$\begin{aligned} L_{G,j} &= 2R_j^{\text{max}}(2N_{<j} + A_{<j} + 12d + 1), \\ C_{\text{ctx},j} &= A_{<j} + N_{<j} + 14d + 2s_0, \\ N_j &= 2(C_{\text{ctx},j} + L_{G,j} + 2s_0), \\ L_{C,j} &= 2R_j^{\text{max}}(2N_j + A_{<j} + 12d + 1), \\ A_j &= L_{C,j} + 2s_0. \end{aligned} \tag{2.1}$$

In particular, A_j is available to the *next* program step, not to the selection of N_j . The coefficients $L_{G,j}$ and $L_{C,j}$ differ because the capture tree includes the new suffix of length n_j , while the program's future tree uses only lower missing lengths.

2.2 Choice of the last scale

Choose an integer κ_D after the preceding finite recursion. Define $\beta_j = u_j \kappa_D / (100p_j R_j^{\text{max}})$, $\tau_j = u_j \kappa_D + L_{\text{occ}} + 2$, and $f = \lceil 4(d+2)(I+1) \rceil$ for each relevant level and tree height. Require

$$\begin{aligned} \kappa_D &> \max\{4 \max_i (N_i, A_i), 100d^2, 4dD\}, \\ 12dN_1/\kappa_D &< d, \quad \kappa_D/3 > 12d^2 + 2, \\ \beta_j &> 2, \quad \tau_j < \kappa_D/4, \quad \kappa_D - \tau_j > 1, \\ (1 - 2c_j)\kappa_D &> L_{\text{occ}} + 5, \\ \kappa_D/4 &> \beta_j + L_{\text{occ}} + 2c_j \kappa_D + 1, \\ (f - I + 1)\kappa_D &> d(I+1)(\kappa_D + 12d) + \tau_j, \\ g_j(c/2)\kappa_D &> d(\kappa_D + 12d) + \tau_j. \end{aligned} \tag{2.2}$$

For the base capture also require $7\kappa_D/8 > L_{\text{occ}} + 5$.

These are finitely many consistent strict inequalities. Indeed $u_j \leq 1/128$ and the coefficient of κ_D in τ_j is u_j . The coefficient in β_j is much smaller than u_j . The last inequality has positive leading margin because $g_j c/2 \geq 2^{d+9}(d+2)$. For the penultimate one, $f \geq 4(d+2)(I+1)$, so its leading coefficient exceeds $d(I+1) + u_j$. All remaining non-leading terms are fixed before κ_D is increased.

Set $C_D = \kappa_D + 10d$, $\lambda_D = 1 + d/\kappa_D$, $a_m = \lfloor \kappa_D \log_2 m \rfloor$, $K = 2^{a_m}$, and $b_m = \lceil C_D \log_2 m \rceil$. Set $B_* = P_* = \lfloor \lambda_D b_m \rfloor$, $S_* = 2^{B_*}$, $Q_* = 2^{P_*}$, $W_* = \lfloor \lambda_D(m+1) \rfloor$, and $w = W_* - 1$. Thus $S_* = Q_*$, and Q_*/K is an integer. Let $\varsigma = \lambda_D C_D$. The estimates used below are

$$\begin{aligned} \varsigma &= \kappa_D + 11d + 10d^2/\kappa_D \leq \kappa_D + 12d, \\ \lambda_D \varsigma - \kappa_D &= 12d + 21d^2/\kappa_D + 10d^3/\kappa_D^2 \leq 14d. \end{aligned} \tag{2.3}$$

Moreover $1 < \lambda_D < 1 + 1/(4D)$.

Fix r, s and put $q = m - r$ and $n_j = \lfloor N_j m / \kappa_D \rfloor$. Eventually $1 \leq n_j < q$, and $N_j > N_{j-1} + d$. Consequently

$$n_j + q - n_{j-1} - w = \frac{N_j - N_{j-1} - d}{\kappa_D} m + O_D(1) > 0. \quad (2.4)$$

All missing suffixes and every slack in this display have positive linear length for fixed D .

2.3 Table indexing and quantifiers

Sample an auxiliary bank of d independent tables $\Pi_1, \dots, \Pi_d$, each mapping $[K] \times [K]^q$ to $[K]$. For each a and every $\mathbf{x} \in [K]^q$, $\Pi_a(\cdot, \mathbf{x})$ is an independent uniform permutation of $[K]$. Separately sample $H_1 : [K]^r \times [K]^q \rightarrow [K]$ with independent uniform entries.

Every assertion G_j or $\mathcal{C}_j$ below is required for every contiguous bank interval $\Pi_a, \dots, \Pi_{a+j-1}$ with $a + j - 1 \leq d$. A first query of length $n \leq q$ means a *fixed raw-prefix slice*: choose $\boldsymbol{\eta} \in [K]^{q-n}$ and use the first table $\Pi_a(\mathbf{k}, \boldsymbol{\eta} \mathbf{x}_1)$. All allowed n and all such prefixes are covered before a competing map is chosen. Overlapping slices do not have to be independent for a union bound.

For the actual component use $H_i = \Pi_{i-1}$ for $2 \leq i \leq s$. The additional auxiliary tables are only a device for stating the induction without an undefined j -fold table family. The iid top step uses $s - 1$ permutation lookups below H_1 .

3 Physical replay systems and their closure

We describe the physical input schedule before introducing any analytical hashes or training experiments. A recurrent run is a nonempty consecutive group of S -layers. Its product state simulates those layers exactly. Throughout the auxiliary arguments there are at most d runs, the product of all their state cardinalities is at most S_*^d , each separating T -layer has radius at most w , and every token belongs to $[Q_*]$.

Definition 3.1 (Physical permutation program). Fix a bank interval of j tables, a first-query length $n \in [n_j, q]$, a fixed first-table prefix, and fixed external data. Draw $\mathbf{k} \sim \text{Unif}([K])$ independently of $\mathbf{x}_1 \in [K]^n$ and $\mathbf{x}_2, \dots, \mathbf{x}_j \in [K]^q$, all uniform. Concatenate the raw queries into a word $\bar{\mathbf{x}}$ of length $L_j(n) = n + (j - 1)q$.

At exactly these physical positions, a preprocessor emits $\mathbf{e}_t = E_t(\bar{\mathbf{x}}^{1:t}) \in [Q_*]$. Its maps may depend on all fixed tables and external data but not on the realized key $\mathbf{k}$. The computation has j recurrent runs $\mathcal{R}_0, \dots, \mathcal{R}_{j-1}$ alternating with $j - 1$ T -layers. Run $\mathcal{R}_0$ starts in an arbitrary state $\phi(\mathbf{k})$. For $1 \leq i < j$, set $t_i(n) = n + iq - n_{j-i}$. Immediately before run $\mathcal{R}_i$ processes position $t_i(n) + 1$, its state can be initialized to $\psi_i(\mathbf{k}, U_0, \bar{\mathbf{x}}^{1:t_i(n)})$, where one common seed U_0 is independent of the key and all raw queries. Thereafter that run follows its fixed update and emission rules.

The output is the last run's output at position $L_j(n)$. The target is the j -fold permutation composition. There are no other input positions, no larger token alphabet, and no extra decoder of the raw future.

The prescribed cuts are separated by more than w eventually, by (2.4). Therefore a local window used after a run's initialization only uses predecessor outputs that are already defined. Behavior before a run's initialization need not be specified.

A capture statement also has already-processed ancestor runs. The following diagonal-state definition makes their role exact.

Definition 3.2 (Capture replay system). Fix $j \geq 1$, $a \geq 0$ with $a + j \leq d$, and a network of $a + j$ recurrent runs $\mathcal{R}_0, \dots, \mathcal{R}_{a+j-1}$ separated by local layers. Fix their maps and a common old underlying token word $\mathbf{c}$ at positions $-aw + 1, \dots, 0$; when $a = 0$ this word is empty.

Let U be a finite parameter family of size $\exp(O_D(m \log m))$, with any probability or subprobability law μ . Independently draw $\mathbf{z} \sim \text{Unif}(F)$, where $|F| = L \leq K$. For each $u \in U$, fix a selector $\xi(u) \in [K]^{q-n_j}$ and a map $f_u : F \rightarrow [K]$ of maximum fiber size at most d_{occ} , where $L/d_{\text{occ}}^2 \geq K/m^{L_{\text{occ}}}$.

The fresh raw queries are $\mathbf{y}_1 \in [K]^{n_j}$ and $\mathbf{y}_2, \dots, \mathbf{y}_j \in [K]^q$, uniform and independent of $(u, \mathbf{z})$. Their underlying tokens are $\mathbf{e}_t = E_t(u, \bar{\mathbf{y}}^{1:t}) \in [Q_*]$, independent of the realized $\mathbf{z}$.

For $0 \leq i \leq a$, run $\mathcal{R}_i$ receives an arbitrary state $\Gamma_i(u, \mathbf{z})$ at time $-(a-i)w$ and follows its rules afterward. Thus the state of the first unresolved run $\mathcal{R}_a$ is supplied at time zero. The tuple of these states is the *early row*; its alphabet is a subproduct of the total state alphabet. At time $n_j + \ell q - n_{j-\ell}$, each later unresolved run $\mathcal{R}_{a+\ell}$, $1 \leq \ell < j$, receives an arbitrary jointly causal state depending on $u, \mathbf{z}$, a shared independent seed, and only the raw prefix already arrived.

The target is the composition of j bank tables starting at $f_u(\mathbf{z})$, with first query $\xi(u)\mathbf{y}_1$ and later queries $\mathbf{y}_2, \dots, \mathbf{y}_j$. A validity mask may reduce success but never changes the raw future distribution. The fixed network rules are common across u ; dependence on u enters through the underlying tokens, early row, and permitted grants.

The parameter-family exponent is fixed before m . For each fixed exponent, the statements are uniform over families within that bound. In the applications here one common exponent is fixed from the depth, resource, and training-tree constants before m and the rival are chosen. Physical old words and a fixed-depth sampled tree have only $O_D(m)$ coordinates in polynomial alphabets, so this condition holds. It is a condition on realized parameter values, not on the description lengths of the competing functions.

Lemma 3.3 (Diagonal reconstruction and rebasing). *A system in Definition 3.2 is well-defined from its old word, early row, and future underlying tokens. Fixing its strict ancestor row $\Gamma_0, \dots, \Gamma_{a-1}$ makes the input to its first unresolved run a source-independent causal word with one Q_* -valued token per original future position.*

Suppose $j > 1$, and set $p = q - n_{j-1}$ and $\ell = n_j + p$. Fix an old parameter u and the first missing suffix $\mathbf{y}_1$, and expose the next prefix $\mathbf{P} \in [K]^p$. Conditional on the underlying segment $\mathbf{v} = \mathbf{e}_{1:\ell}$, the remaining system is a capture replay system with $j-1$ unresolved lookups and $a+1$ ancestor runs. Its source parameter is $\mathbf{P}$ with its actual posterior law, and $\mathbf{z}$ remains independent uniform on F .

Proof. Run $\mathcal{R}_0$ starts at $-aw$. The first output required from the next run occurs after $-(a-1)w$, and its local window begins strictly after $-aw$. Inducting over runs and positions reconstructs all required outputs. For fixed strict ancestor states these computations are functions of u and the arrived raw future only. Their output alphabet is still $[Q_*]$.

At the new cut ℓ , give ancestor run i its actual state at $\ell - (a+1-i)w$, for $0 \leq i \leq a$, and give run $a+1$ its actual grant at ℓ . For each old ancestor, this is later than its old initialization by $\ell - w > 0$. Its state is therefore defined by the already processed physical history. The new old word is the last $(a+1)w$ underlying tokens ending at ℓ . It is determined by the old fixed word and $\mathbf{v}$. Conditional on $\mathbf{v}$, this word is common across the posterior values of $\mathbf{P}$.

The new early row is not conditioned on: it remains an arbitrary function of $(\mathbf{P}, \mathbf{z})$. Since $\mathbf{v}$ is independent of the realized $\mathbf{z}$, conditioning on it preserves uniformity and independence of $\mathbf{z}$. The new source profile is the old profile followed by the now-completed permutation, so occupancy is unchanged. Subtracting ℓ from every later cut gives $n_{j-1} + iq - n_{(j-1)-i}$, the prescribed lower schedule. The old state-product bound is unchanged. $\square$

Remark 3.4 (What is and is not being granted). A common old token word is fixed in a capture application, while the early row stays random under the source law. Fixing the row is legitimate for a pointwise replay calculation, but not for claiming a uniform conditional source. Analytical

tags introduced later are not physical input tokens. None is made available at an earlier physical position merely because it is useful in the analysis.

4 Uniform finite random-table estimates

Every statement in this section first fixes a candidate, proves a probability bound for that candidate, and then takes a finite union over *all* candidates of the declared sizes. Accordingly the resulting event also covers candidates chosen after the tables. Counting realized contexts alone would not establish this quantifier.

4.1 One recurrent scan

Lemma 4.1 (Uniform single-scan bound). *With probability $1 - o(1)$ over the permutation bank, every physical one-run program of Definition 3.1, for every first-table slice of length $n \in [n_1, q]$, has success at most $m^{-\kappa_D/16}$. The input encoding may even assign an arbitrary noncausal word in $[Q_*]^n$ to each query column.*

Independently, the analogous bound holds for the iid table H_1 , uniformly over every $F \subseteq [K]^r$ with $K/m^{12d^2} \leq |F| \leq K$, arbitrary initial maps on F , and q -token word assignments.

Proof. Pad the run state set to cardinality $H = S_*^d$. Fix an initial map, its position-dependent transitions, and the final emission. Put $\theta = \frac{1}{2}m^{-\kappa_D/16}$ and $h = \lceil \theta K \rceil$. One physical word determines a response vector indexed by the K input keys. For a uniform permutation column, a prescribed injective set of h responses has probability $1/(K)_h$, and a noninjective one has probability zero. Hence

$$\Pr[\text{at least } h \text{ agreements}] \leq \binom{K}{h} \frac{1}{(K)_h} = \frac{1}{h!}. \quad (4.1)$$

Union over the Q_*^n possible words, allowing the maximizing word to be selected after the column is known. For the fixed machine template, one column is bad with probability at most $Q_*^n/h!$. Different columns are independent. If $J_{\text{bad}} = \lceil m^E \rceil$, then

$$\Pr[\text{at least } J_{\text{bad}} \text{ bad columns}] \leq \binom{K^n}{J_{\text{bad}}} (Q_*^n/h!)^{J_{\text{bad}}} \leq \exp[-\Omega(J_{\text{bad}} h \log m)]. \quad (4.2)$$

Here $h = m^{15\kappa_D/16+o(1)}$, so $\log h!$ dominates both $n \log Q_*$ and $n \log K$.

The number of templates has logarithm at most

$$K \log H + nH Q_* \log H + H \log K = m^{O_d(\kappa_D)}. \quad (4.3)$$

This counts the initial map, all transitions, and the output map; allowing emissions at other positions only adds a term of the same polynomial form. Choose the fixed exponent E large enough to dominate (4.3). There are at most $O(m)K^q$ fixed prefix/length slices, whose logarithm is $O_D(m \log m)$. Their union is absorbed as well. On the event obtained, success is at most $h/K + J_{\text{bad}}/K^n \leq m^{-\kappa_D/16}$ for sufficiently large m .

For the iid table and a fixed source set of size L , use $h_F = \lceil \theta L \rceil$. Agreement on h_F source points has probability at most

$$\binom{L}{h_F} K^{-h_F} \leq (e/(\theta K))^{h_F}. \quad (4.4)$$

The word union adds Q_*^q . Since $L \geq K/m^{12d^2}$, $h_F \gg m$. The same exceptional-column argument applies. In addition, all choices of F cost at most 2^{K^r} , whose logarithm is polynomial in m .

Increase E to absorb it. Initial maps on a fiber have at most $H^L \leq H^K$ possibilities, already covered by the template envelope. This proves the iid assertion without treating iid columns as permutations. $\square$

4.2 Iid occupancy and balance

Lemma 4.2 (All-fiber occupancy). *With probability $1 - o(1)$, simultaneously for every $F \subseteq [K]^r$ with $|F| \leq K$, all but an m^{-100d} fraction of first-query columns have maximum fiber size at most m^2 for $H_1(\cdot, \mathbf{x})|_F$.*

Proof. For a fixed F and column,

$$\Pr \left[\max_{\mathbf{k}} |\{\mathbf{z} \in F : H_1(\mathbf{z}, \mathbf{x}) = \mathbf{k}\}| \geq m^2 \right] \leq K \binom{|F|}{m^2} K^{-m^2} \leq K(e/m^2)^{m^2}. \quad (4.5)$$

The K^q column events are independent. A binomial tail for at least $m^{-100d} K^q$ failures has negative logarithm of order at least $m^{-100d} K^q$ for large m . This dominates the logarithm $K^r \log 2$ of all source subsets. A union over those subsets proves the claim. The claim bounds a fraction of columns separately for each F ; it is not a pointwise assertion for all pairs $(F, \mathbf{x})$. $\square$

Lemma 4.3 (Iid first-table balance). *Let $n = \Theta_D(m)$, let H have independent uniform $[K]$ -valued entries on $[K]^r \times [K]^n$, and put $M = K^n$. Let $\delta = m^{-\kappa_D/3}$. Suppose $|F| = L \geq K/m^{12d^2}$, $t \leq n - \eta m$ for a fixed $\eta > 0$, and $\gamma : F \times [K]^t \rightarrow [P_0]$, $e : [K]^n \rightarrow [J]$, where $P_0 \leq S_*^d$ and $J \leq m^{\chi m + o(m)}$. If $\chi < \liminf n \log_m K/m$, then, on one event of probability $1 - o(1)$, all these choices satisfy*

$$\text{TV}(\mathcal{L}(C, H(\mathbf{z}, \mathbf{x})), \mathcal{L}(C) \otimes \text{Unif}([K])) \leq \delta, \quad C = (\gamma(\mathbf{z}, \mathbf{x}^{1:t}), e(\mathbf{x})), \quad (4.6)$$

under independent uniform $\mathbf{z} \in F$ and $\mathbf{x} \in [K]^n$. The event includes all eligible source subsets and the fixed finite collection of slice types used below.

Proof. Fix the subset and maps. Let N_c be the number of input cells in context c , and let $T_{c,k}$ count those with output k . The averaged conditional variation is $(2LM)^{-1} \sum_{c,k} |T_{c,k} - N_c/K|$. Independence gives $\text{Var}(T_{c,k}) \leq N_c/K$. Cauchy–Schwarz and at most $P_0 J$ contexts give expected variation at most $\frac{1}{2} \sqrt{K P_0 J / (LM)}$. The strict leading entropy gap makes this exponentially smaller than δ .

Changing one entry changes the variation by at most $1/(LM)$. Bounded differences therefore gives failure at most $\exp[-c\delta^2 LM]$ for an absolute $c > 0$. The maps have at most $P_0^{LK^t} J^M$ possibilities. Their logarithms are $LK^t \log P_0$ and $M \log J$. The former is smaller by the positive linear suffix gap. The latter is dominated because

$$\delta^2 L \geq m^{\kappa_D/3 - 12d^2 + o(1)} \gg m \log m. \quad (4.7)$$

All subset choices add at most $K^r \log 2$, and fixed raw-prefix slice choices add $O_D(m \log m)$ to the logarithm. These are absorbed. Since the union enumerates all maps, their eventual table dependence causes no problem. $\square$

4.3 Permutation balance at a short first query

Lemma 4.4 (Finite permutation balance). *Let $\{\pi_{\mathbf{x}} : \mathbf{x} \in [K]^n\}$ be independent uniform permutations, put $M = K^n$, and let $0 \leq t < n$. For fixed alphabet sizes P_0, J and $\delta > 0$ satisfying*

$P_0 J \leq \delta^2 M$, the probability that some maps $\gamma : [K] \times [K]^t \rightarrow [P_0]$ and $e : [K]^n \rightarrow [J]$ violate $\text{TV}(\mathcal{L}(C, \pi_{\mathbf{x}}(\mathbf{k})), \mathcal{L}(C) \otimes \text{Unif}([K])) \leq \delta$ is at most

$$\exp\{K^{t+1} \log P_0 + M \log J - \delta^2 K M / 32\}, \quad C = (\gamma(\mathbf{k}, \mathbf{x}^{1:t}), e(\mathbf{x})), \quad (4.8)$$

where the input key and query are independent uniform.

Proof. For fixed maps and context c , let $b_{c,\mathbf{x}}$ be the number of keys in that context at column $\mathbf{x}$, and $N_c = \sum_{\mathbf{x}} b_{c,\mathbf{x}}$. For a fixed output key, the number X_c of occurrences is a sum of independent Bernoulli variables with means $b_{c,\mathbf{x}}/K$. Thus $\mathbb{E}X_c = N_c/K$ and $\text{Var}(X_c) \leq N_c/K$. The exact expected variation and its bound are

$$\mathbb{E} \text{TV} = \frac{1}{2M} \sum_c \mathbb{E}|X_c - N_c/K| \leq \frac{1}{2M} \sqrt{P_0 J \sum_c N_c/K} = \frac{1}{2} \sqrt{P_0 J/M}. \quad (4.9)$$

This expression is an upper bound, not an equality in general.

A swap of two output images in one column changes at most two input cells and changes the variation by at most $2/(KM)$. Expose individual images within each permutation. Completions under two possible next images can be coupled by swapping those two unused images. Each of the KM Doob martingale differences is therefore at most $2/(KM)$ in absolute value. Azuma gives an upper tail $\exp(-v^2 KM/8)$ at deviation v . Under the hypothesis, the mean is at most $\delta/2$; take $v = \delta/2$. Finally there are $P_0^{K^{t+1}} J^M$ pairs of maps. Unioning proves (4.8). $\square$

For the scales of Section 2, use $\delta = m^{-\kappa_D/3}$ and $P_0 \leq S_*^d$. At a query of length n_j , a label rate $\chi < N_j$ ensures $P_0 J \leq \delta^2 K^{n_j}$. If $n - t = \Theta_D(m)$, division of the exponent in (4.8) by M gives

$$K^{1-(n-t)} \log P_0 + \log J - \delta^2 K / 32 = -m^{\kappa_D/3+o(1)} / 32 + O_D(m \log m) < 0. \quad (4.10)$$

All fixed raw-prefix slices and allowed lengths are covered by one union. The condition for the shortest query is $\chi < N_j$, not merely $\chi < \kappa_D$. For example, if a label stores the entire short query and an early state stores the input key, the conditional output is a point mass for every table.

4.4 A finite capture bound on a smaller source

Lemma 4.5 (Finite independent-column capture estimate). *Let $A, L_y, N, D_0, J, d_0, K, L$ be positive integers with $N \leq A$ and $\theta L > ed_0^2$. Sample independent uniform permutations $\pi_{a,y}$ of $[K]$ for $(a, y) \in [A] \times [L_y]$. Define*

$$\begin{aligned} C &= N \log A + NL(\log K + \log D_0) + NL_y \log J + D_0 J \log K, \\ H &= (\theta N L_y L / d_0) \log(\theta L / (ed_0^2)). \end{aligned} \quad (4.11)$$

Except with probability $\exp(C - H)$, all choices of distinct $a_1, \dots, a_N$, maps $f_u : [L] \rightarrow [K]$ of occupancy at most d_0 , maps $e_u : [L] \rightarrow [D_0]$, messages $\phi_u : [L_y] \rightarrow [J]$, and a common decoder $g : [D_0] \times [J] \rightarrow [K]$ have $\Pr[g(e_u(\mathbf{z}), \phi_u(y)) = \pi_{a_u,y}(f_u(\mathbf{z}))] < \theta$ under independent uniform $u \in [N]$, $\mathbf{z} \in [L]$, and $y \in [L_y]$.

Proof. Fix a candidate. In column (u, y) , a live key has at most d_0 demanded output values because its preimage under f_u has at most d_0 source points. Let $B_{u,y}$ count keys whose permutation image lies in its demanded set, and $M_{u,y}$ count successful source points. Then $M_{u,y} \leq d_0 B_{u,y}$ and

$$\mathbb{E} \binom{B_{u,y}}{h} \leq \binom{K}{h} \frac{d_0^h}{(K)_h} = \frac{d_0^h}{h!}. \quad (4.12)$$

Expanding $(1 + v)^{B_{u,y}}$ yields $\mathbb{E}e^{\lambda M_{u,y}/d_0} \leq \exp[d_0(e^\lambda - 1)]$ for $\lambda \geq 0$. The NL_y columns are independent. Take $e^\lambda = \theta L/d_0^2$ and apply the exponential Markov inequality. The probability of at least $\theta NL_y L$ successes is at most e^{-H} . The choices counted in $\mathbb{C}$ are, respectively, selectors, profiles and early maps, messages, and the common decoder. Unioning proves the lemma. $\square$

The same statement allows a fixed outer permutation library applied after $\pi_{a_u,y}$, by adding $NL_y \log M_{\text{out}}$ to $\mathbb{C}$ and applying the inverse outer permutation to each demanded output. Only the version above is needed in the present base case. Relabeling a source fiber by $[L]$ causes no uncounted choice: all profiles and early maps were already enumerated.

4.5 Signatures and prefix response sets

Lemma 4.6 (Uniform pair signatures). *If a permutation table has at most M columns, the probability that two distinct columns agree on at least h keys is at most $M^2/h!$. Consequently all final-table column pairs in this proof agree on at most $\tau_{\text{sig}}K$ keys, with probability $1 - o(1)$, where $\tau_{\text{sig}} = 2m^{-\kappa_D/4}$.*

Proof. The relative permutation of two independent columns is uniform. A union over h fixed points gives $\binom{K}{h}/(K)_h = 1/h!$. Union over pairs proves the first claim. Use $h = \lceil m^{-\kappa_D/4}K \rceil$, $M = K^q$, and $3\kappa_D/4 > 1$. The factor two absorbs rounding. The same proof applies to distinct fixed-length composed column words by conditioning on all but one column at their latest differing stage, although that stronger version is not needed for the tail argument below. $\square$

Lemma 4.7 (Uniform small-response-set prefix bound). *Let $\pi_{\xi,y}$, $(\xi, y) \in [A] \times [L_y]$, be independent uniform permutations. Fix integers a, g, n, N with $g \leq a$ and $n \leq K/2$, and a row alphabet of size D_0 . For distinct roots $y_1, \dots, y_a$ and sets $B_{v,i} \subseteq [K]$ of size at most αK , call a prefix ξ good if at least n keys $\mathbf{k}$ admit a row v and at least g indices i such that $\pi_{\xi,y_i}(\mathbf{k}) \in B_{v,i}$. Put*

$$T = \binom{K}{n} \left[D_0 \binom{a}{g} (2\alpha)^g \right]^n. \quad (4.13)$$

The probability that some root tuple and set array have N distinct good prefixes is at most $L_y^a 2^{aD_0K} A^N T^N$.

Proof. For fixed n keys choose a row and a g -subset for each key. Reveal the required images key by key. In each column at least $K - n \geq K/2$ images remain, of which at most αK belong to the desired set. Thus every required membership has conditional probability at most 2α . This proves the bound T for a fixed prefix. Different prefixes use independent columns. There are at most L_y^a root tuples, 2^{aD_0K} set arrays, and A^N ordered prefix selections. Their union gives the claim, including post-table choices of the response sets. $\square$

If $K = m^{\kappa_D + o(1)}$, $D_0 \leq m^{d_\zeta + o(1)}$, $\alpha = m^{-e + o(1)}$, and $n = \lfloor K/m^\tau \rfloor$, the conditions $ge > d_\zeta + \tau$ and $\kappa_D - \tau > 1$ make $\log T \leq -c'n \log m$ for some fixed $c' > 0$. Choosing a sufficiently large fixed polynomial $N = m^{E_{\text{cap}} + o(1)}$, with $E_{\text{cap}} > d_\zeta + \tau$, defeats the set-array count and all root/prefix choices. The conclusion is a polynomial list of good prefixes. If the whole prefix alphabet is smaller than this polynomial, the conclusion is simply the trivial full list. This convention avoids any vacuous choice of $N > A$.

4.6 Terminal tuple dictionaries and a finite cylinder cover

A cylinder in $[M]^R$ fixes selected coordinates to specified values and leaves all other coordinates free.

Lemma 4.8 (Tuple event with arbitrary parent inputs). *Partition R terminal coordinates among at most A_0 parents. Let $\rho_\zeta, \zeta \in [M]$, be independent uniform permutations of $[K]$, and let a dictionary $\mathcal{W} \subseteq [K]^R$ have size at most D_{row} . An injective selector tuple $(\zeta_1, \dots, \zeta_R)$ is n -good if there exist dictionary rows $\mathbf{w}_1, \dots, \mathbf{w}_n$ and, for each parent p , distinct inputs $k_{p,1}, \dots, k_{p,n}$ such that $\rho_{\zeta_i}(k_{p,t}) = w_{t,i}$ at every coordinate i belonging to parent p .*

For $1 \leq f \leq R$, put $T_{\text{par}} = D_{\text{row}}^n / [n!(K)_n^{f-A_0}]$. For any integer $J_{\text{pet}} \geq 2$, except with probability at most

$$\exp[O(RD_{\text{row}} \log K + RJ_{\text{pet}} \log M + RJ_{\text{pet}} \log(R+1))] T_{\text{par}}^{J_{\text{pet}}}, \quad (4.14)$$

every such dictionary's good selector tuples are covered by at most $[R^2 J_{\text{pet}}]^{R-f+1}$ cylinders, each fixing $R - f + 1$ coordinates.

Proof. Consider tuples with a common coordinate-wise core and with disjoint sets of noncore labels between tuples. Suppose every tuple has at least f noncore coordinates. Condition on all core permutation columns and select f fresh columns in one tuple. There are at most $(K)_n^{A_0} D_{\text{row}}^n$ choices of ordered parent input arrays and dictionary rows. Simultaneously permuting the n sample indices leaves the constraint event unchanged and acts freely, because every parent array is injective. Dividing by $n!$ is therefore valid.

Each fresh column imposes n distinct-input constraints. Repeated prescribed outputs make the event impossible; otherwise its probability is $(K)_n^{-1}$. Fresh columns are independent. Thus the conditional probability for one tuple is at most T_{par} . Different tuples in this family have disjoint fresh columns, so the bound multiplies. Union over dictionaries, tuples, and coordinate-core placements gives (4.14). Parent inputs may be selected after all tables and may differ between parents; they have been explicitly counted.

On the complement event no family of J_{pet} such tuples exists. We now give the covering argument. Start with no coordinates fixed. At a node with fewer than $R - f + 1$ fixed coordinates, take a maximal family of tuples whose remaining label sets are pairwise disjoint. Its size is less than J_{pet} , because every remaining set has size at least f . The union of its remaining labels has size at most $R(J_{\text{pet}} - 1)$ and intersects the remaining labels of every tuple at that node. Cover by children that fix one additional coordinate to one of these labels. There are at most $R^2 J_{\text{pet}}$ children. Repeat for $R - f + 1$ levels. Injectivity of each tuple ensures its noncore labels do not equal a core label. The leaves are the stated cylinder cover. $\square$

In our application $D_{\text{row}} = S_*^{d(I+1)}$, $A_0 \leq I$, $n = \lfloor K/m^{\tau_j} \rfloor$, and $f = \lceil 4(d+2)(I+1) \rceil$. Stirling's bound gives

$$\log T_{\text{par}} \leq -[(f - I + 1)\kappa_D - d(I+1)\varsigma - \tau_j - o(1)]n \log m. \quad (4.15)$$

The bracket is positive by (2.2). Since $\kappa_D - \tau_j > 1$, a sufficiently large fixed polynomial J_{pet} makes (4.14) tend to zero, including all fixed raw-prefix slice choices. Its cylinder count is polynomial in m because R is fixed before m .

For a noninjective source profile, distinct inputs at *all* terminal parents will be justified by selecting sources with distinct values of the one common initial profile. Each subsequent path applies a permutation to that profile. Bounded occupancy at unrelated parent maps would not by itself justify the hypothesis.

5 Deterministic and probabilistic bookkeeping lemmas

5.1 Balanced reindexing of a finite code

Lemma 5.1 (One common balanced hash). *Let $\mathcal{D}$ have size BJ , and let $\mathcal{A}_1, \dots, \mathcal{A}_F$ be subsets of $\mathcal{D}$, each of size at most B . A uniform random bijection $\text{Hash} : \mathcal{D} \rightarrow [B] \times [J]$ has some first-*

coordinate bucket containing at least b images from some $\mathcal{A}_i$ with probability at most $FB(e/b)^b$.

Proof. For a fixed subset and bucket, the factorial moment of its load is at most $\binom{B}{b}B^{-b} \leq (e/b)^b$. Apply Markov to the binomial coefficient of the load and union over subsets and buckets. $\square$

We use the lemma after fixing the rival and its finite family of raw histories, not as a new restriction on the rival. Here $\log F + \log B = O_D(m \log m)$ and $b = \lceil m^{\beta_j} \rceil$ with $\beta_j > 2$. A suitable common hash therefore exists. One hash is fixed for the whole relevant family before the analytical training tree is sampled. It is not reselected separately for each source or training outcome.

Lemma 5.2 (Exact branch change of variables). *Let $\mathcal{X}$ and $\mathcal{B}$ have the same finite cardinality. Suppose $\iota_\rho : \mathcal{B} \rightarrow \mathcal{X}$ is a partial injection, and each raw point assigned to branch ρ is its image. For every function f on $\mathcal{X}$,*

$$\mathbb{E}_{b \sim \text{Unif}(\mathcal{B})} \mathbf{1}_{\{b \in \text{dom } \iota_\rho\}} f(\iota_\rho(b)) = \mathbb{E}_{x \sim \text{Unif}(\mathcal{X})} \mathbf{1}_{\{x \in \text{Im } \iota_\rho\}} f(x). \quad (5.1)$$

For chronological compositions of such maps, the identity iterates with earlier raw histories held fixed. If a branch's bucket and rank can be recovered from the raw history by the time of a frontier choice, any policy on the branch pulls back to a legal raw-history policy at that frontier. Invalid histories can be assigned arbitrary default states.

Proof. Both sides sum exactly once over the image, with denominator $|\mathcal{X}| = |\mathcal{B}|$. Iterate this equality by conditional expectation. At a prescribed raw cut, compute only forward hashes whose entire defining segment has already arrived, then use the branch policy's chosen state. This preserves the joint choices at all later cuts and preserves a shared independent random seed. No law is conditioned on validity; validity remains a payoff multiplier. $\square$

5.2 Thin final code fibers

Lemma 5.3 (Successful mass in thick tail fibers). *Let $\mathbf{z}$ be uniform on a set F of size L , and let $f : F \rightarrow [K]$ have occupancy at most d_0 . Let final targets $T_\eta(\mathbf{z}) = \rho_\eta(f(\mathbf{z}))$ have pairwise agreement on at most $d_0 \tau K$ source points. Suppose a source-independent tail code $B(\eta)$ and all already-fixed state data determine the prediction, so the prediction at each $\mathbf{z}$ is constant on a code fiber. The successful mass contributed by code fibers of size greater than b is at most*

$$\frac{1}{b+1} + \frac{b}{2} \frac{d_0 \tau K}{L}. \quad (5.2)$$

This remains an upper bound with any additional validity mask.

Proof. In any $b+1$ distinct members of one fiber, let $a(\mathbf{z})$ be the number of correct predictions. Since the prediction is common, $a \leq 1 + \binom{a}{2}$. Average over $\mathbf{z}$ and use the pairwise agreement bound. Dividing by $b+1$ gives (5.2). For a larger fiber, average over uniform $(b+1)$ -subsets; each member has the same inclusion probability. Then average over fibers with their original weights. A mask only removes successes. $\square$

The requirement that every frontier state already be fixed before the tail is essential. Before that point, equal underlying codewords need not imply equal later granted states. We use Lemma 5.3 only at the final missing suffix, after the last grant.

5.3 Training moments and diffuse selectors

Lemma 5.4 (Shared-prefix training moment). *Fix a source parameter u , an input key/source point, and an actual causal policy. Let its branch-masked success probability on one future path be $p(u, \mathbf{z})$. Sample a regular b -ary tree of height h : each node samples independent child continuations conditional on the ancestors, and the state's choice at a node is shared by its children. With $R = b^h$, the probability that the actual policy succeeds at every leaf is at least $p(u, \mathbf{z})^R$.*

Consequently, if $a_T(u)$ is the fraction of source points correct on all training leaves and p is the original branch-success mass under a probability or subprobability measure μ on u , then $\mathbb{E}_T \int a_T(u) d\mu(u) \geq p^R$.

Proof. At height one the children are independent conditional on the root data, giving the b th power. At greater height, apply the induction to each child subtree and then Jensen's inequality to the random child segment. Iterating gives power b^h . Finally apply Jensen over u and the uniform source point. For a subprobability measure, the inequality $(\int f d\mu)^R \leq \int f^R d\mu$ follows by adding one zero-payoff dummy point or by Holder's inequality. All branch-validity indicators remain in the leaf payoff. $\square$

Lemma 5.5 (Diffuse posterior capture). *Let $\xi : U \rightarrow \mathcal{X}$ be a selector and let μ be a subprobability measure with $\mu(\xi = x) \leq 1/N$ for all x . Suppose a training experiment satisfies the moment conclusion of Lemma 5.4. Suppose also that for each training draw T there are a bad-source set B_T and a list $L_T \subseteq \mathcal{X}$ such that $\mathbb{E}_T \mu(B_T) \leq \eta$, $|L_T| \leq M$, and every $u \notin B_T$ with $a_T(u) \geq \theta$ has $\xi(u) \in L_T$. Then*

$$p^R \leq \theta + \eta + M/N. \quad (5.3)$$

For a fixed source-independent old code context c , define $\mathcal{P}(c) = \{x : \mu_c(\xi = x) > 1/A\}$ using its actual posterior law. Then $|\mathcal{P}(c)| < A$. If the preceding hypotheses hold under restriction to the complementary selector values, without renormalizing, their successful mass is at most $(\theta + \eta + M/A)^{1/R}$.

Proof. For each T , split the integral of a_T into training-light sources, bad sources, and the rest. These contributions are at most θ , $\mu(B_T)$, and $\mu(\xi \in L_T) \leq M/N$, respectively. Average and apply the moment inequality. For the posterior statement, remove the heavy selector atoms and keep all other original weights. The resulting measure has atom cap $1/A$ and total mass at most one. Apply the first assertion and then average over old contexts with their actual probabilities. No independence between the selected lists and the tables or training draw is used. $\square$

A sourcewise bound $\Pr_T[u \in B_T] \leq \eta$ ensures that the same bad-mass bound survives every pre-future restriction of μ . Future-dependent masks cannot be incorporated into μ ; they stay inside the trial indicators.

5.4 Causal tests and their false-pass probabilities

Lemma 5.6 (Nested causal tester). *At depth i of a finite future experiment, a policy chooses a state in an alphabet of size H_i after seeing the current raw history v and before the next random segment. Let the optimal value be $V_i(v) = \max_s \mathbb{E}[V_{i+1}(v, s, X)]$, with a $\{0, 1\}$ terminal payoff. At a sampled node with b_i children, declare a pass if some one state makes at least g_i children*

pass their recursive tests. If β_i bounds false passing at nodes with $V_i \leq \delta_i$, then

$$\begin{aligned}\beta_i &\leq H_i \binom{b_i}{g_i} (\delta_i / \delta_{i+1} + \beta_{i+1})^{g_i}, \\ \beta_{\text{last}} &\leq H_{\text{last}} \binom{b_{\text{last}}}{g_{\text{last}}} \delta_{\text{last}}^{g_{\text{last}}}.\end{aligned}\tag{5.4}$$

The same state is used for all selected children of a node.

Proof. Fix a candidate state. Markov's inequality bounds the fraction of children with true value greater than δ_{i+1} by δ_i / δ_{i+1} . At any other child the false-pass probability is at most β_{i+1} . Child sample trees are independent for this fixed state. Union over g_i -subsets and then over the H_i possible states. The terminal formula is the same argument for literal Bernoulli success. No independence between tests for different states is required. $\square$

For a level- j proof use $b_i = b_j^{\text{tr}}$, $g_i = g_j$, and $\delta_i = m^{-z_i \kappa_D}$ with $z_i = c/2^{i+1}$, where $c = \min_{\ell < j} c_\ell$. Since $H_i \leq S_*^d$ and the height is at most d , the smallest threshold difference has exponent at least $c/2^d$. The choice of g_j makes $g_j c/2^d \geq 1024(d+2)$. Backward induction in (5.4) gives $\beta_i \leq m^{-100(d+2)\kappa_D}$ eventually. Union over the fixed number of nodes and at most $S_*^d K$ strict ancestor-row/input-key values leaves a sourcewise bad probability at most $m^{-90(d+2)\kappa_D}$.

This argument first conditions on *that source's own* sampled ancestor history and then averages. It does not union over all old sources, all raw histories, or alignment-prefix arrays selected later. The root threshold is $\delta_0 = m^{-c\kappa_D/2}$; it is not a fixed constant independent of m .

Lemma 5.7 (Fixed terminal coordinates force a test subtree). *In a full b -ary tree of height h , a set of fixed leaves contains a recursive g -of- b subtree unless it omits at least $(b - g + 1)^h$ leaves. For $b = 2g_j$ and the f of Section 2, one has $f < (g_j + 1)^h$. Hence every cylinder fixing all but at most $f - 1$ terminal coordinates supplies such a subtree, chosen by a deterministic rule depending on the cylinder only.*

Proof. A failed internal node has at least $b - g + 1$ failed children. Induct on height and add their disjoint sets of missing leaves. For the numerical claim, $I + 1 \leq 2R/(2g_j - 1)$ and $(g_j + 1)^h \geq R/2^d$. The choice $g_j \geq 2^{d+10}(d+2)/c$ is much larger than required for $\lceil 4(d+2)(I+1) \rceil < R/2^d$. Ties between qualifying subtrees can be resolved lexicographically, independently of the realized source point. $\square$

6 Chronological normalization of a replay

In this section assume the lower capture assertions $\mathcal{C}_i$, $i < j$, with list sizes at most $A_i^*(m) = \lceil m^{A_i m} \rceil$ and errors $\epsilon_i = m^{-c_i \kappa_D}$. We establish the normalization needed at level j without changing a physical position or a frontier's information set.

6.1 An intermediate missing suffix and next prefix

At a node with $i > 1$ unresolved lookups, the next raw segment consists of a missing suffix $\mathbf{y} \in [K]^{n_i}$ and an arrived next-query prefix $\mathbf{P} \in [K]^p$, where $p = q - n_{i-1}$. Hold the old parameter and $\mathbf{y}$ fixed. The underlying code segment $\mathbf{V}_u(\mathbf{y}, \mathbf{P})$ has exactly $n_i + p$ tokens. Lemma 3.3 permits application of $\mathcal{C}_{i-1}$ after conditioning on this segment. The next prefix has its actual posterior law, while the source point remains uniform and the early row remains unconditioned.

Let $\mathcal{P}_{u,\mathbf{y}}(\mathbf{V})$ be the resulting list of prefixes. Removing successful inputs with $\mathbf{P}$ outside this list costs at most ϵ_{i-1} under the original raw law. On retained inputs append the ordinal of $\mathbf{P}$ in

that list to the code. For each fixed $(u, \mathbf{y})$, the map $\mathbf{P} \mapsto (\mathbf{V}_u(\mathbf{y}, \mathbf{P}), \text{ord}_{u, \mathbf{y}}(\mathbf{P}))$ is injective on its retained domain.

Choose one common bijection from the code/ordinal alphabet to $[K]^p \times [J_i]$, where

$$J_i = Q_*^{n_i} (Q_*/K)^{q-n_i-1} A_{i-1}^*(m). \quad (6.1)$$

It is an integer. By Lemma 5.1, the first coordinate, called a bucket, has at most $b = \lceil m^{\beta_j} \rceil$ valid preimages for every member of the finite old-parameter/raw-history family. An inverse rank in $[b]$ selects a partial injection from the bucket space to the raw prefix space. The second coordinate is the *residual tag*. A bucket and residual tag determine the original underlying code segment, because the hash is a bijection on the augmented code alphabet.

The raw suffix $\mathbf{y}$ is sampled separately and identically across sources in a training experiment. The completed segment's hash is evaluated only at its right-hand cut, after all coordinates of $(\mathbf{y}, \mathbf{P})$ have arrived. An earlier physical computation is never allowed to read that bucket, ordinal, or inverse rank. Normalized buckets are analytical random variables, not substitute early physical tokens.

6.2 The last missing suffix

At the final suffix of length n_1 , all grants have already arrived. For fixed old parameter and history, write the underlying tail code as $\mathbf{B}_u(\boldsymbol{\eta}) \in [Q_*]^{n_1}$. A fixed source point's prediction is constant on its code fibers, because all other state data precede $\boldsymbol{\eta}$. Lemmas 4.6 and 5.3 bound the successful mass in fibers larger than b by

$$\frac{1}{b+1} + \frac{b}{2} \frac{d_{\text{occ}} \tau_{\text{sig}} K}{L} \leq O(m^{-\beta_j}) + O(m^{\beta_j - \kappa_D/4 + L_{\text{occ}}}). \quad (6.2)$$

The profile at this point is a permutation of f_u , so its occupancy is unchanged.

For the remaining fibers, hash the code alphabet bijectively to $[K]^{n_1} \times [J_{\text{tail}}]$, where $J_{\text{tail}} = (Q_*/K)^{n_1}$. Each fixed old-history image contains at most K^{n_1} different codes. A common balanced hash gives at most b code images per bucket. One rank selects the code, and another rank selects one of its at most b raw preimages. Thus there are at most b^2 tail inverse ranks. Bucket and tag again determine the original code.

6.3 What the normalized experiment preserves

Proposition 6.1 (Normalization contract). *For a level- j capture system, or the $(j-1)$ -lookup future in a level- j program proof, the preceding operations give the following properties.*

1. *Apart from the sum of lower capture errors on one raw path and the final thickness bound, raw successful mass partitions among at most $b^{j+1} \leq m^{p_j \beta_j + o(1)}$ branch vectors. A vector uses one prefix inverse rank per depth and the two tail ranks.*
2. *On any fixed branch, successive raw suffixes and normalized buckets retain their independent uniform laws before validity masks. The analytical-to-raw map is a chronological partial injection. Equation (5.1) preserves masked success exactly. The same branch rank at a depth is used at every training node of that depth.*
3. *All lower capture lists, hashes, ordinal maps, and partial inverses are selected before training. They are not changed by later prefix alignment. Their values depend on the old parameter and arrived raw history, never on the realized source point.*
4. *A complete vector of residual tags for a fixed training tree determines the underlying physical codeword on every sampled edge, once the shared raw suffixes, sampled buckets, and fixed old context are given. This determination is common to all source parameters with that tag vector.*

5. The tag-vector alphabet has size at most $m^{L_{C,j}m+o(m)}$ for a capture tree and at most $m^{L_{G,j}m+o(m)}$ for a program's future tree.

Proof. The first four conclusions follow by iterating the two constructions and Lemma 5.2. To charge lower capture errors, first hold the previously arrived raw history and current shared suffix fixed, apply the uniform lower capture estimate to the next raw prefix, and then average. Other masks can be dropped when upper-bounding this loss. There are at most $j - 1$ such applications on a fresh path; no tree-size or branch-count multiplier is introduced at this step. Branches are introduced only after these raw losses are charged.

For the last claim, (2.3) and (6.1) give tag rate at most $2N_i + 12d + A_{i-1}$ at an intermediate node. The tail tag has rate at most $12d$. There are at most $2R_j^{\max}$ sampled nodes/edges contributing tags. In a capture tree $i \leq j$; in the program future $i \leq j - 1$. Equation (2.1) gives the two asserted bounds. Polynomial ordinal/refinement factors contribute $o(m)$ to the base- m exponent. The inverse branch ranks are charged separately to the probability calculation.

For hash existence, the entire family of old parameters and possible raw ancestors at these fixed depths has $\exp(O_D(m \log m))$ realized values. Lemma 5.1 applies with $\beta_j > 2$. No union over descriptions of all competing encoders is needed to choose an analytical hash after one encoder has been fixed; uniformity of the hard tables is provided by the separate finite events of Section 4. $\square$

Pointwise lower-program legality. At any normalized node, fix the prototype source parameter, its own arrived raw history, the common old code, and its *strict* ancestor row. Lemma 3.3 reconstructs one key-independent causal Q_* -valued input token at each remaining physical position. The current run's initial state and later frontier states may still be optimized. Pulling a normalized policy back by the prototype's forward hashes, only at their declared cuts, gives a physical program of Definition 3.1. No extra positions, wider alphabet, or lookahead is introduced. This is the legality statement used below for response sets; equality of a final code by itself would not have sufficed before all grants arrived.

7 Training dictionaries and bottom-up alignment

This section supplies the structural implication behind the moment estimates. All tables satisfy the finite events in Section 4, all lower G_i and C_i used here hold, and one physical system and its normalization maps have been fixed. None of those maps is reselected after observing a training sample.

7.1 The training experiment and its common dictionary

Fix a normalized inverse-branch vector. Use the regular b_j^{tr} -ary tree of the relevant height h . At an internal node with more than one lookup left, a child draw consists of the shared raw missing suffix and a next-prefix bucket. The next frontier is chosen after that entire draw and is shared by that child's descendants. At the final level, draw independent tail buckets. Conditional on ancestors, all child draws and their descendant sample trees are independent. This is precisely the chronology required by Lemmas 5.4 and 5.6.

For an old parameter u , every valid sampled path has an actual underlying codeword and actual frontier states. Group parameters by their complete tag fingerprint. Proposition 6.1 makes the underlying codes common on every sampled edge of one group. To describe every possible vector of its R final outputs, allow one complete initial early row and one state for each subsequent internal-node frontier. Each coordinate is bounded by the physical path-state

product S_*^d . Therefore one common dictionary of terminal output vectors has size at most

$$D_{\text{tree}} \leq S_*^{d(I+1)}. \quad (7.1)$$

This is an upper bound on the whole tree, not just one path. It allows different states at different internal nodes, but not different states for children sharing a node. Outputs outside $[K]$ can be replaced by a fixed element of $[K]$ when forming a containing dictionary; this cannot remove any genuinely correct output.

Remove parameters for which the sampled valid terminal-column selectors collide. Conditional on a fixed source and earlier ancestors, a tail inverse is injective on a domain of size K^{n_1} . For two independently sampled tails, equality of the valid full selectors therefore has probability at most K^{-n_1} . The same bound follows after fixing different parent histories; different final raw prefixes make equality impossible. A union over the R leaves costs at most $\binom{R}{2} K^{-n_1}$. Repeated shared raw suffixes among siblings have the analogous bound. These are sourcewise probabilities: averaging them does not introduce a factor equal to the parameter-family size.

Call u training-heavy if at least $\theta_j L$ source points are correct at every training leaf, including all validity masks, where $\theta_j = m^{-u_j \kappa_D}$. Among these points, at least $\theta_j L / d_{\text{occ}}$ distinct values of f_u occur. Since $L / d_{\text{occ}}^2 \geq K / m^{L_{\text{occ}}}$ and $d_{\text{occ}} \geq 1$, one can choose $n_{\text{key}} = \lfloor K / m^{L_j} \rfloor$ source points with distinct f_u values. The same chosen points have distinct target inputs at every terminal parent: each path applies a permutation to the common initial profile. Thus Lemma 4.8 applies even for a noninjective initial profile.

For each fingerprint, its heavy selector tuples are covered by polynomially many terminal cylinders. By Lemma 5.7, each cylinder provides a canonical recursive g_j -of- b_j^{tr} subtree of fixed terminal coordinates. Its shape is independent of the realized source point. Assign a tuple belonging to several cylinders to the first one in a fixed ordering.

7.2 Own tests and small sets of free input keys

For each prospective prototype u , node, its own arrived raw history, strict ancestor-state row g , and free input key $\mathbf{k}$, define $q_{u,g}(\mathbf{k})$ to be its best fresh continuation value on its own normalized branch. Optimize the tested run's initial state and every subsequent legal causal frontier state. The payoff includes the prototype's original inverse-branch masks. The strict ancestor row does not include the state of the tested run. That state is optimized; its actual value in the global dictionary can only witness an admissible choice.

These values and their recursive tests are defined before the descendant training draws. Later grouping only chooses which preexisting prototype and value are used. In particular, a value is never redefined against an alignment-prefix vector supplied by a different parameter.

Lemma 7.1 (A single program realizes the keywise optima). *Fix a prototype, its arrived raw prefix and strict ancestor row, and suppose i lookups remain. Its own values satisfy $K^{-1} \sum_{\mathbf{k}} q_{u,g}(\mathbf{k}) \leq \epsilon_i$ whenever the lower assertion G_i applies. Consequently $|\{\mathbf{k} : q_{u,g}(\mathbf{k}) > \delta\}| \leq (\epsilon_i / \delta) K$.*

Proof. Finite state and raw-history alphabets give a deterministic maximizing policy by backward recursion. Its state at a node is chosen before the next raw segment, not independently for different future children. Choose a tie-broken maximizing policy for each input key. Because the key is an allowed argument of every frontier map, these policies combine into one initial map and one jointly causal family of later maps on the disjoint key-indexed histories. Randomization cannot improve the maximum of a scalar expected payoff.

By the pointwise legality conclusion of Section 6, this is one physical i -lookup program on the original remaining positions. If the values were defined in bucket coordinates, pull the policy back by the prototype's own forward hashes at the corresponding arrived-prefix cuts. Lemma 5.2

preserves its masked value. Dropping the mask only increases the success bounded by G_i . This gives the average bound, and Markov's inequality gives the response-set bound. $\square$

The thresholds used by the own tests are $\delta_\ell = m^{-c\kappa_D/2^{\ell+1}}$ from Section 5. At every invoked lower continuation, $\epsilon_i \leq m^{-c\kappa_D}$, so its response-set fraction is at most $m^{-c\kappa_D/2}$. Call a parameter own-bad if one of its sampled tests at any required node falsely passes when its own true value is below its threshold, for some strict ancestor row and free input key. The sourcewise bound following Lemma 5.6 is at most $\eta_{\text{own},j} = m^{-90(d+2)\kappa_D}$. Define and remove these bad parameters before choosing representatives. This costs at most $\eta_{\text{own},j}$ in expected parameter mass. No union over parameter values or subsequently chosen alignment-list ranks is needed.

7.3 The alignment implication

Lemma 7.2 (Bottom-up own-prefix alignment). *Fix a training sample and one inverse branch. After the sourcewise own-bad and collision deletions, every training-heavy parameter in a fingerprint/cylinder group can be classified by polynomially many additional source-independent rank fields so that its required target prefixes agree with those of a representative of its refined group on the canonical test subtree. For a capture system, its initial selector belongs to a polynomial list attached to the final group. For a program's remaining future, whose initial arrived prefix is already fixed, every retained all-training-correct source point passes the selected representative's own full future test at that point's actual incoming live key.*

Proof. At the bottom, the cylinder has fixed the complete final-table selectors on the canonical subtree. In particular their arrived final-query prefixes agree across parameters in the group. The fingerprint also fixes all underlying codewords on the sampled paths.

Proceed upwards. Assume all target prefixes strictly below the current level agree throughout a group at the retained nodes. Choose a representative that survived its own badness tests; all current parameters have this property. Consider an actual all-training-correct source point from any parameter in the group. At a child continuation, use its actual incoming key as a free key argument of the representative, and use its actual strict ancestor row. Three facts show that the representative's own sampled test passes. First, the common fingerprint reconstructs the same encoded words. Second, the already aligned lower prefixes and fixed terminal selectors make the remaining target compositions on the sampled subtree identical. Third, the actual frontier states witness admissible state choices, with the same state used by all children of a node. These are the representative's own sampled raw-prefix paths below that child, not foreign prefix assignments. The representative's own-goodness therefore makes its fresh optimized value exceed the threshold.

The row in the resulting response-set comparison is common across qualifying siblings. At their parent, collect the strict ancestor states and the current run's state, one coordinate per physical run. Its alphabet is at most S_*^d . Given that row and the common codeword on a child edge, diagonal reconstruction determines the child's strict ancestor row. The tested child's own state remains optimized. Thus the child response set can be written as $B_{g,i}$ for one parent row g and child index i . Independent row choices for different siblings are not being granted or counted.

Lemma 7.1 bounds each such set by αK , where $\alpha \leq m^{-c\kappa_D/2}$. Across qualifying siblings, the actual child input keys have the form $\Pi(\mathbf{k}, \mathbf{P}(u)\mathbf{y}_i)$: the parent key $\mathbf{k}$ and parent row are the same, the shared suffixes $\mathbf{y}_i$ are distinct, and $\mathbf{P}(u)$ is the current target prefix. Training heaviness supplies at least n_{key} distinct parent keys. Lemma 4.7, with the last margin in (2.2), therefore puts $\mathbf{P}(u)$ in a polynomial list determined by the group, node, and chosen representative. Its uniform event covers response sets selected after the tables.

At each retained node of this level, refine the group by the actual list rank of its prefix. There are

only a fixed number of nodes; the product of their polynomial rank alphabets is still polynomial in m . The refined groups now have matching prefixes at this level as well. Choose a new representative inside a refined group for the next level. Every earlier agreement is a group-membership condition on actual prefixes, so replacing the representative preserves it. No lower capture map, hash, own value, or badness definition changes.

Repeat to the top. For capture, the current selector is the original arrived prefix, and the final response-set argument supplies its list. For a program future, that first remaining prefix was fixed externally; the same transfer at the root yields the representative's own full future test. The refinements depend on the parameter and the fixed training sample, not on the particular source point. This proves both conclusions. $\square$

Proposition 7.3 (Finite capture and program certificates). *Under the preceding lower-stage hypotheses, the normalized experiments have the following certificates.*

For capture, if the selector marginal of a parameter subprobability measure has atoms at most $1/N$, then for each inverse branch with successful mass p_b ,

$$p_b^R \leq \theta_j + \eta_{\text{own},j} + \eta_{\text{coll},j} + M_j/N, \quad M_j \leq m^{L_{C,j}m+o(m)}. \quad (7.2)$$

Here $\eta_{\text{coll},j}$ is exponentially small in $m \log m$.

For a program's future, after fixing its common old code and initial arrived prefix, there is an additional source-independent label of size at most $m^{L_{G,j}m+o(m)}$. For each fixed training sample it selects a representative and an indicator I_T of retained all-training-correct source points. Writing $H_T = \mathbb{E}I_T$ under the original source and parameter weights, one has

$$\mathbb{E}_T H_T \geq p_b^R - \theta_j - \eta_{\text{own},j} - \eta_{\text{coll},j}. \quad (7.3)$$

At every retained source point, the representative's own optimized future value, with its strict early row and that point's actual live key, exceeds $\delta_0 = m^{-c\kappa_D/2}$.

Proof. For capture, union the initial-selector lists over fingerprints, terminal cylinders, and bottom-up refinement groups. The cylinder counts and all refinement/list counts are fixed polynomials in m . Only the fingerprint contributes a leading $m \log m$ rate, bounded by $L_{C,j}$. Lemma 7.2 provides the heavy-source implication required in Lemma 5.5. The training moment, sourcewise bad deletions, and collision estimate give (7.2).

For the program future, apply the same grouping using its shorter tree, whose fingerprint rate is $L_{G,j}$. Retain the heavy, nonbad, noncolliding, all-training-correct points. Training-light parameters contribute at most θ_j before averaging. Lemma 5.4 proves (7.3), and Lemma 7.2 gives the pointwise own-value implication. Labels and representative selection use the fixed training data and old parameter only. A whole representative is lookup data indexed by that label, not an extra label field containing its raw query coordinates. $\square$

8 The mutual program/capture induction

For every bank interval of length j , let G_j be the assertion that every program in Definition 3.1 has success at most $\epsilon_j = m^{-c_j \kappa_D}$. Let $\mathcal{C}_j$ assert that every system in Definition 3.2 has an analytical list of initial selectors of size at most $A_j^*(m) = \lceil m^{A_j m} \rceil$, outside which successful mass is at most ϵ_j . This list depends on the fixed old context, parameter law, and system, not on the realized source point or fresh future.

We first fix every input-independent frontier seed and prove the deterministic statements. The final capture lists are the heavy selector atoms of the given posterior parameter law; they do not

depend on the frontier seed. Thus the same lists satisfy the averaged capture bounds after the seed is restored. Program bounds are uniform for each seed and also average directly.

All primitive events are chosen on one table realization. Their finite estimates include all bank intervals, declared suffix lengths, and fixed first-query slices. For fixed D there are only $\exp(O_D(m \log m))$ such raw-prefix slices, and the adjustable polynomial exceptional-column, petal, and prefix-list degrees give failure probabilities that dominate this extra union. The finite lemmas are universal over their displayed candidate maps and dictionaries. The additional unions over integer source cardinalities $L \leq K$, allowable occupancy bounds, and state-alphabet sizes cost only fixed polynomial factors and are included as well. Their intersection has probability $1 - o(1)$. The induction that follows is deterministic on this intersection, including subsequent choices of rivals, analytical hashes, and representatives.

8.1 The two different base assertions

The assertion G_1 is Lemma 4.1. We prove $\mathcal{C}_1$ separately, retaining its source-size and occupancy parameters.

Fix its old context. After all early states have arrived, its output is a common function of the early row and complete underlying suffix code. In Lemma 4.5, take the row alphabet at most $D_0 = S_*^d$, message alphabet $J = Q_*^{n_1}$, suffix count $L_y = K^{n_1}$, and a sample of $N = \lceil m^{dm} \rceil$ distinct selectors. Set the agreement tolerance to $\theta = m^{-\kappa_D/8}$. The source condition gives $\theta L/d_{\text{occ}}^2 \geq m^{7\kappa_D/8 - L_{\text{occ}} + o(1)}$. This dominates every per-column polynomial cost. The decoder count is dominated as well: its leading exponential-in- m rate is $\zeta N_1/\kappa_D$, strictly below $N_1 + d$ because $12dN_1/\kappa_D < d$. Thus the finite bound tends to zero with the required uniform margin. All common replay decoders are included by allowing every map on the displayed row/message alphabets.

It follows that any law whose selector atoms are at most $1/N$ has success at most θ . To justify the passage from uniform distinct-selector samples, the probability vectors on a finite selector alphabet with atom cap $1/N$ form the convex hull of the uniform laws on N -element subsets. A vertex cannot have two coordinates strictly between zero and $1/N$, since they could be perturbed in opposite directions; the sum constraint then forces every nonzero vertex coordinate to equal $1/N$. For each selected selector, sample its conditional parameter and apply the universal finite bound before averaging those choices.

Let $A = A_1^*(m)$ and remove selector atoms larger than $1/A$. Suppose the remaining original mass is β . If $\beta < N/A$, its successful mass is at most N/A . Otherwise normalize this remaining measure: its selector atoms are at most $1/(A\beta) \leq 1/N$, so its success before undoing normalization is at most θ . In either case the unnormalized successful mass is at most $\theta + N/A$. Since $A_1 = 2d$, this is eventually at most $m^{-\kappa_D/16}$. The same conclusion holds for a subprobability law by the identical mass split. The capture list has fewer than $A_1^*(m)$ elements. This proves $\mathcal{C}_1$.

8.2 Same-cut incidence and the complete first-query label

Assume the assertions for every smaller stage count. Fix a j -lookup program, $j \geq 2$, on a bank interval starting at index ν , and an incoming length $n \in [n_j, q]$. Write its first query as $\mathbf{A}$ and its second query as $(\boldsymbol{\xi}, \mathbf{Y}_1)$, where $\boldsymbol{\xi} \in [K]^p$, $p = q - n_{j-1}$, and $\mathbf{Y}_1 \in [K]^{n_{j-1}}$. Put $c = n + p$ and $t = c - w$. Equation (2.4) gives $0 < t < n$ and a positive linear unseen suffix $n - t$.

The old underlying word at this cut is $\mathbf{v}(\mathbf{A}, \boldsymbol{\xi}) = \mathbf{e}_{t+1:c} \in [Q_*]^w$. The state of run $\mathcal{R}_0$ at t has the form $\Gamma = \gamma(\mathbf{k}, \mathbf{A}^{1:t})$. The grant to run $\mathcal{R}_1$ at c is not included in this balance row; it can depend on the complete first query and $\boldsymbol{\xi}$.

Apply $\mathcal{C}_{j-1}$ to parameter $u = (\mathbf{A}, \boldsymbol{\xi})$, with source point $\mathbf{k}$ and profile $f_u(\mathbf{k}) = \Pi_\nu(\mathbf{k}, \boldsymbol{\eta}\mathbf{A})$, where

η is the externally fixed first-table prefix. Condition only on the underlying word $\mathbf{v}$. This is a capture system with one ancestor: its diagonal early states are the state of $\mathcal{R}_0$ at t and the grant of $\mathcal{R}_1$ at c . They remain unconditioned under the uniform source law. Since $\mathbf{v}$ is source-independent and precedes the remaining future, the posterior law of $(\mathbf{A}, \xi)$ is independent of the uniform source and fresh future. Definition 3.2 and Lemma 3.3 apply exactly.

Consequently there is a list $\mathcal{P}(\mathbf{v})$ of at most $A^* = \max_{i < j} A_i^*(m)$ possible prefixes, and the unconditional successful mass not captured by this list is at most ϵ_{j-1} . For each prefix define $\mathcal{B}_\xi = \{\mathbf{v} : \xi \in \mathcal{P}(\mathbf{v})\}$. Double counting and the original uniform prefix law give

$$\sum_{\xi} |\mathcal{B}_\xi| \leq A^* Q_*^w, \quad \Pr[|\mathcal{B}_\xi| > L_{\text{ctx}}] \leq \frac{A^* Q_*^w}{K^p L_{\text{ctx}}}. \quad (8.1)$$

Take $L_{\text{ctx}} = \lceil m^{C_{\text{ctx},j}m} \rceil$. By (2.3), the right-hand side is at most $m^{-(2s_0 - o(1))m}$.

Fix a nonexceptional ξ . On captured first queries, record the rank of $\mathbf{v}(\mathbf{A}, \xi)$ in $\mathcal{B}_\xi$; use a sentinel elsewhere. This is a total source-independent map of the full first query $\mathbf{A}$. It does not store the raw suffix $\mathbf{A}^{t+1:n}$ or the externally fixed ξ .

For each fixed old-word rank, apply the program-future certificate in Proposition 7.3 to the remaining $j - 1$ lookups. Parameter $\mathbf{A}$ has its actual posterior within the rank, and the original source key remains independent and uniform. All future normalization and own tests are fixed before their training draw. The complete label $e(\mathbf{A})$ consists of the rank, future fingerprint, terminal-cylinder index, polynomial alignment-list ranks, and validity flags. A sentinel covers every excluded first query. Its alphabet satisfies

$$|\text{Im}(e)| \leq m^{(C_{\text{ctx},j} + L_{G,j})m + o(m)}, \quad C_{\text{ctx},j} + L_{G,j} < N_j. \quad (8.2)$$

Training may be chosen separately for different old-word ranks and then fixed collectively; weighted Jensen gives the same global training moment. This costs a product probability space, not a product label alphabet. A realized query records only its own rank and its own group's fingerprint. The simultaneous balance event already ranges over all resulting total maps e .

8.3 The optimized crossed continuation

Fix a nonexceptional prefix, an inverse branch, all normalization maps, and the complete analytical training draw. For each nonsentinel label a , let $\mathbf{A}_a$ be its chosen full-query representative and let $\mathbf{v}_a$ be its common old underlying word. For fixed label a , strict early row g , and free key $\mathbf{k}'$, let $q_{a,g}(\mathbf{k}')$ be the representative's own optimized remaining continuation value, including its own original branch masks. Give the sentinel value zero.

There is a direct physical construction behind this value. Start the old run $\mathcal{R}_0$ in state g at time t , feed $\mathbf{v}_a$ through time c , and thereafter feed the representative's underlying tokens at the same original positions. The local window at $c + 1$ starts at $t + 1$. Thus this simulation supplies exactly one source-independent causal token at each position of the first remaining run. Its rules and tape are fixed by (a, g) and the fresh raw future; no latent actual first query can change them. Optimize the first remaining run's state at c and all later grants. Their shifted cut times are exactly the lower program's times, since

$$t_{i+1}(n) - c = n_{j-1} + iq - n_{(j-1)-i}. \quad (8.3)$$

Lemma 7.1 therefore gives $K^{-1} \sum_{\mathbf{k}'} q_{a,g}(\mathbf{k}') \leq \epsilon_{j-1}$ for each fixed (a, g) . The program has the original remaining state product and no added physical positions.

Let $\mathbf{K}_{\text{live}} = \Pi_\nu(\mathbf{k}, \eta \mathbf{A})$ under the original source/first-query law. The program certificate gives a retained indicator I_T such that $I_T = 1$ implies $q_{e(\mathbf{A}), \Gamma}(\mathbf{K}_{\text{live}}) > \delta_0$, where $\delta_0 = m^{-c\kappa_D/2}$. With

$$H_T = \mathbb{E} I_T,$$

$$\begin{aligned} \delta_0 H_T &\leq \mathbb{E}[I_T q_{e,\Gamma}(\mathbf{K}_{\text{live}})] \leq \mathbb{E} q_{e,\Gamma}(\mathbf{K}_{\text{live}}) \\ &\leq \epsilon_{j-1} + \mathbb{E}_{e,\Gamma} \text{TV}(\mathcal{L}(\mathbf{K}_{\text{live}} \mid e, \Gamma), \text{Unif}([K])) \leq \epsilon_{j-1} + \delta_{\text{TV}}, \end{aligned} \quad (8.4)$$

where $\delta_{\text{TV}} = m^{-\kappa_D/3}$. The last inequality is the simultaneous permutation-balance event of Lemma 4.4, applied to the complete total label (8.2) and the early map at t . The raw first-query slice has length at least n_j , and its label rate is strictly below N_j . The expectation keeps the original weights, before any conditioning on retained success, a group, or a representative. No number-of-labels factor multiplies the total-variation error.

Averaging (8.4) over the training draw and using (7.3) gives, for each fixed prefix and inverse branch,

$$p_b^R \leq \frac{\epsilon_{j-1} + \delta_{\text{TV}}}{\delta_0} + \theta_j + \eta_{\text{own},j} + \eta_{\text{coll},j}. \quad (8.5)$$

This uses an optimizing causal policy as an analytical certificate. It does not claim that the actual sampled states attain the same value on a fresh future.

8.4 Quantitative closure of both assertions

The right-hand side of (8.5) is $O(m^{-u_j \kappa_D})$. Indeed $\epsilon_{j-1}/\delta_0 \leq m^{-c\kappa_D/2}$, $\delta_{\text{TV}}/\delta_0 = m^{-(1/3-c/2)\kappa_D}$, and $u_j = c/8$. Summing the R th roots over all inverse branches costs at most

$$m^{p_j \beta_j + o(1)} [O(m^{-u_j \kappa_D})]^{1/R} \leq m^{-[99u_j/(100R_j^{\max}) + o(1)]\kappa_D}. \quad (8.6)$$

The lower capture errors were charged on a raw path before this sum and are $O(m^{-c\kappa_D})$. The exceptional-prefix incidence error is exponentially small. The final thickness error (6.2) has exponent greater than $2c_j \kappa_D$, using (2.2); in particular $\beta_j = 4c_j \kappa_D$. These terms together are eventually at most $m^{-c_j \kappa_D}$. This proves G_j uniformly over the allowed first-query lengths and slices.

For $\mathcal{C}_j$, fix its old context and remove all selector atoms larger than $1/A_j^*(m)$. This is a source-independent list with fewer than $A_j^*(m)$ elements. Keep the original weights of all other parameters. On this restricted subprobability measure, Proposition 7.3 gives (7.2) with $N = A_j^*(m)$, and

$$M_j/A_j^*(m) \leq m^{-(A_j - L_{\mathcal{C},j})m + o(m)} = m^{-2s_0 m + o(m)}. \quad (8.7)$$

Take branch roots and add the normalization losses exactly as above. The successful mass outside the list is at most $m^{-c_j \kappa_D}$. All sourcewise badness estimates survive the restriction because they were proved separately for each parameter before its measure was specified. Future validity masks remain inside the trials. Average over old contexts with their actual weights when such contexts are random. This proves $\mathcal{C}_j$.

The induction is well-founded. The program step uses only lower capture and program assertions, and its fingerprint uses only lower N_i, A_i . The new N_j is already fixed when the capture tree and its rate $L_{\mathcal{C},j}$ are used. Neither a finite rate inequality nor the abstract tester is being substituted for the physical normalization and alignment lemmas proved above.

9 The iid first stage and arbitrary source chunks

Set $L_0 = \lfloor K^r / (m^{10d} S_*^{r-1}) \rfloor$. Since $r + s \leq D$, the exponent $10d + (r-1)(\varsigma - \kappa_D)$ is strictly smaller than $12d^2$. Floors and ceilings change only bounded factors. Hence, for all sufficiently large m , one has $K/m^{12d^2} \leq L_0 \leq K/2$ and $m^2/L_0 = o(1)$.

Theorem 9.1 (Uniform iid-source random-chain bound). *There is a realization of H_1 and the permutation bank, fixed before all competing maps, such that the following holds simultaneously for every $F \subseteq [K]^r$ with $L_0 \leq |F| \leq K$. Draw $\mathbf{z}_0 \sim \text{Unif}(F)$ and independent uniform $\mathbf{x}_1, \dots, \mathbf{x}_s \in [K]^q$. Define $\mathbf{z}_1 = H_1(\mathbf{z}_0, \mathbf{x}_1)$ and $\mathbf{z}_i = H_i(\mathbf{z}_{i-1}, \mathbf{x}_i)$, with $H_i = \Pi_{i-1}$ for $i \geq 2$.*

Outside an m^{-100d} fraction of first queries, the map $\mathbf{z}_0 \mapsto \mathbf{z}_1$ on F has occupancy at most m^2 . Moreover, every physical s -stage replay of Definition 3.1, with source $\mathbf{z}_0$ in place of its initial key, full first-query length q , and the iid-first-stage target above, has success at most

$$\Pr[\hat{\mathbf{z}} = \mathbf{z}_s] \leq m^{-c_s \kappa_D} + o(1). \quad (9.1)$$

The $o(1)$ is uniform over source chunks, fixed old contexts, and post-table replay maps. In particular, it includes the unconditional exceptional first-column mass and need not have exponent $c_s \kappa_D$.

Proof. Intersect the bank event proved in Section 8 with the uniform iid occupancy, balance, and single-scan events of Section 4. Each has probability $1 - o(1)$; their intersection therefore has positive probability. Fix such a realization. The occupancy assertion is Lemma 4.2. For $s = 1$, the replay is one run with state size at most S_*^d , and the iid part of Lemma 4.1 proves (9.1) directly.

Suppose $s \geq 2$. We repeat the first-stage certificate construction of the G_s proof, not its permutation-first-stage theorem. The incoming first query has full length q . Its first later cut is $c = 2q - n_{s-1}$ and its balance frontier is $t = c - w$, with $0 < t < q$. The actual early map is $\Gamma = \gamma(\mathbf{z}_0, \mathbf{x}_1^{1:t})$.

First remove the source-independent bad first-column set from the successful mass, at cost at most m^{-100d} . On the remaining columns, each parameter $\mathbf{x}_1$ gives a profile $f_{\mathbf{x}_1}(\mathbf{z}_0) = H_1(\mathbf{z}_0, \mathbf{x}_1)$ of occupancy at most m^2 . All later target maps are permutations. Therefore

$$\frac{|F|}{(m^2)^2} \geq \frac{K}{m^{12d^2+4}} \geq \frac{K}{m^{L_{\text{occ}}}}. \quad (9.2)$$

This is the source condition of Definition 3.2. Retaining the source-independent good-column flag as part of the total label permits every balance estimate below to remain under the original full first-query law.

Apply $\mathcal{C}_{s-1}$ at the second-query cut. The source parameter consists of the first query and the arrived second-query prefix; the source point remains uniform on F . Conditional on the underlying old word, that parameter has its actual posterior, and the early row stays unconditioned. The rebasing and same-cut incidence arguments are unchanged. In particular, the old-word inverse-context rate is at most $C_{\text{ctx},s}$.

Next perform the future normalization, training, and bottom-up alignment from Sections 6–7. Their uses of source uniformity concern the original source point in F , not a purported uniform iid output. For a training-heavy parameter, the distinct-key count is at least $|F|m^{-u_s \kappa_D}/m^2 \geq \lfloor K/m^{\tau_s} \rfloor$. The same selected initial profile values are distinct at all terminal parents, since every subsequent map is a permutation. Thus the finite source-size, tuple, and prefix-incidence bounds apply with no extra per-level occupancy loss.

On each fixed second-prefix slice, the complete source-independent label has rate at most $C_{\text{ctx},s} + L_{G,s} < N_s < \kappa_D$. The representative's own future tape and maps are fixed by that label and the early row. Its optimized root and later causal policies paste, key by key, into one G_{s-1} program on the same remaining positions. Lemma 7.1 bounds their mean value on a uniform live key by ϵ_{s-1} .

Only the first balance step changes. Apply Lemma 4.3 to the original law of $\mathbf{z}_0 \in F$ and the full first query. It is uniform over the complete table-adaptive label and the early map missing the

positive linear suffix $q - t$. Therefore the analogue of (8.4) is

$$\delta_0 H_T \leq \text{Cross}_T^{\text{opt}} \leq \epsilon_{s-1} + m^{-\kappa_D/3}. \quad (9.3)$$

All excluded columns and inputs are represented by sentinels or validity indicators; the expectation still has the original weights. The training moment, branch summation, and direct-loss calculation of Section 8 now prove (9.1). Finally add the unconditional bad-column mass. Every invoked table event precedes the choice of F , its posterior parameter laws, and all replay maps. $\square$

Actual latent states versus the optimizing certificate. One can also represent the conditional old data, given the complete label, early row, and live key, using one independent seed. This preserves joint causality of actual later grants; Appendix A gives the precise statement. That simulation is not the source of the lower bound in (9.3). The lower bound comes from the own-tested maximizing policy, whose legality has been proved separately.

10 The task, the architectural reduction, and completeness

We prove Theorem 1.1. Fix D , all constants, and one table realization from Theorem 9.1. The constants are common to every allowed r, s ; finitely many components can use different table realizations.

10.1 The two tagged branches

On the chain branch draw $\mathbf{z}_0 \sim \text{Unif}([K]^r)$ and independent uniform $\mathbf{x}_1, \dots, \mathbf{x}_s \in [K]^q$. Its input is $I = (\mathbf{z}_0, \mathbf{0}^{4D(m+1)}, \mathbf{0}^r, \mathbf{x}_1, \dots, \mathbf{x}_s)$. The first zero block is the guard and the second is the scratch block. The target is $\mathbf{z}_s$, defined by the iid-first-stage chain in Theorem 9.1. Its length is $L_{\text{ch}} = 2r + 4D(m+1) + sq$.

On the reach branch let $k_0 = \lfloor m/2 \rfloor$. Draw independent uniform $\mathbf{u}_1, \dots, \mathbf{u}_{k_0} \in [K]$ and an independent uniform $J \in [k_0]$. In a tape of length $sm + 1$, place the database at its first k_0 positions and J at its last position, with blanks elsewhere. The target is $\mathbf{u}_J$. Left-pad this tape to length L_{ch} . Both branches carry a branch tag, type tags, and position information in their tokens; all their alphabets have size $\text{poly}_D(m)$. Choose each branch with probability $1/2$. The task domain is the set of these formatted inputs. A total function on all length- L_{ch} words can be obtained by using the completeness model's output on every other input; this does not change the task on the support of its input law.

10.2 An exact target model within the stated resources

On the chain branch, number the target recurrent layers from bottom to top by $1, \dots, r$. Layer i copies its input until the source position containing coordinate $z_{0,r-i+1}$. It retains this coordinate, emits prescribed padding until scratch position $r - i + 1$, releases the retained coordinate at that position, and then resumes copying. At any release position, every higher layer has already released its own coordinate and is copying. Thus the top recurrent output has the original source vector in the scratch block, in the correct order, followed by unchanged raw queries.

A layer never needs two independent full-width payload registers: before capture and after release its register holds the token being copied, whereas between capture and release it holds only its retained source coordinate. A position counter and a fixed number of mode/type/branch indicators accompany that register. The state and output types therefore have cardinality at most Km^{c_0} for an absolute fixed $c_0 < 10d$ and all sufficiently large m ; the constant number of types can depend on D . They fit in b_m , because $b_m - a_m = 10d \log_2 m + O(1)$.

At the endpoint of $\mathbf{x}_1$, the first T-layer reads the r scratch coordinates and the q query symbols. There are $r + q = m$ such positions, within its window $m + 1$. It writes $\mathbf{z}_1$ at that endpoint and preserves all future query positions. At each subsequent query endpoint, the next T-layer reads the preceding endpoint and q current symbols, and writes the next key. Its window is at most $q + 1 \leq m + 1$. Earlier local layers preserve the later query blocks needed for these evaluations. The final output is $\mathbf{z}_s$.

On the reach branch, the recurrent layers copy. Each of the first $s - 1$ local layers moves the database to the right by m positions while preserving the query token at the final position. The final layer sees the shifted bank and J in one window of size $m + 1$: after all shifts the first bank coordinate is at relative position $1 + (s - 1)m$, exactly m positions before the query. It outputs $\mathbf{u}_J$.

Use the branch tag to select these layerwise rules. The state is a tagged disjoint union of the two state types, not a parallel product of their states. The fixed overhead is included in the same reserve. Hence one model of profile $S^r T^s$ solves the mixture exactly with the required budgets.

10.3 The forward moving-cut lemma

Lemma 10.1 (Guard reconstruction). *Consider a prefix of a depth-at-most- D competing architecture, containing h recurrent layers. On the chain input, there is a source-only map $\Gamma : [K]^r \rightarrow \mathcal{G}$, with $|\mathcal{G}| \leq S_*^h$, such that after fixing $\Gamma(\mathbf{z}_0)$, all prefix-output tokens on the original scratch and query positions are common source-independent causal functions of the raw queries.*

Proof. Start with a cut immediately after the r source coordinates. When crossing a recurrent layer, record its state at the current cut and keep the cut fixed. When crossing a local layer, advance the cut by w . Inductively, recorded states reconstruct every subsequent recurrent output from already reconstructed inputs, and a local window after the advanced cut lies entirely in the reconstructed portion of its input.

There are at most D local layers, and $Dw < 4D(m + 1)$. Every recorded cut lies inside the guard, so each recorded state is a function of the source alone. Their tuple has at most S_*^h values. All scratch and query positions lie after the final cut. Once the tuple is fixed, their outputs are functions only of arrived raw query symbols and the tuple; they cannot depend on the particular source point in its fiber. $\square$

Let $F_\gamma = \Gamma^{-1}(\gamma)$. Whenever $h \leq r - 1$, the total uniform-source mass in fibers of size smaller than L_0 is at most

$$\frac{S_*^{r-1} L_0}{K^r} \leq m^{-10d}. \quad (10.1)$$

Partition each other fiber into disjoint chunks with sizes in $[L_0, 2L_0)$. Such a partition is obtained by making blocks of size L_0 and distributing the remainder among them. Conditional on a chunk, the source is uniform. Every chunk has size at most K and is covered by the same table event.

10.4 Competitors with too few local layers

Suppose A has at most $s - 1$ T-layers. Work on the reach branch and place the initial backward cut immediately before J . Moving backwards across a recurrent layer records its state at the current cut. Moving backwards across a local layer shifts the cut left by at most w . Each recorded state is before J and hence independent of J . At the bottom, the cut is to the right of the database, because in the unpadded coordinates

$$sm - (s - 1)w = (s - (s - 1)\lambda_D + o(1))m > m/2. \quad (10.2)$$

Inductive replay from these states and the known remaining blanks and query reconstructs the prediction. Thus it is a function of J and a message $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_{k_0})$ of at most DB_* bits.

Independence of the database coordinates and entropy subadditivity give $\sum_i I(\mathbf{u}_i; \Gamma) \leq H(\Gamma) \leq DB_*$. Since J is independent uniform, $I(\mathbf{u}_J; \Gamma, J) \leq DB_*/k_0$. If prediction success is p , its error indicator gives $H(\mathbf{u}_J | \Gamma, J) \leq 1 + (1 - p) \log_2 K$. Consequently

$$p \leq \frac{I(\mathbf{u}_J; \Gamma, J) + 1}{\log_2 K} \leq \frac{DB_*/k_0 + 1}{\log_2 K} = o(1). \quad (10.3)$$

The fixed left padding affects none of these relative distances.

10.5 Competitors with too few recurrent layers

Suppose A has fewer than r S-layers. Apply Lemma 10.1 to the whole model, discard the small fibers in (10.1), and condition on one retained chunk F . For fixed queries its final prediction is constant in the source point. Except on an m^{-100d} fraction of first columns, the first target map has fibers of size at most m^2 on F . Subsequent permutations preserve these fibers. Therefore

$$\Pr[\text{chain success}] \leq m^{-10d} + m^{-100d} + \frac{m^2}{L_0} = o(1). \quad (10.4)$$

This branch needs only occupancy, not the multistage replay bound.

10.6 All remaining avoiding profiles

Now suppose A has at least r recurrent layers and s local layers, but avoids $S^r T^s$. Cut the architecture immediately before its r th recurrent layer. The strict prefix has $r - 1$ recurrent layers. Lemma 10.1 and the chunking argument make its scratch/query output source-independent on each retained chunk F .

Write the residual profile as $S^{b_0} T S^{b_1} \dots T S^{b_j}$, with $b_0 \geq 1$ and $b_i \geq 0$. Avoidance implies $j \leq s - 1$. The number $B_{\text{res}} = \sum_i b_i$ of residual recurrent layers satisfies $B_{\text{res}} \leq D - r - s + 1$, since the full architecture contains at least s local layers.

Merge every nonempty recurrent run into its product state. Insert identity local layers until there are $s - 1$ local layers, and put a copying recurrent layer in every empty recurrent slot. There are at most $s - 1$ such insertions, because the first slot is nonempty. A copying recurrent layer stores and emits its current token and therefore uses $Q_* = S_*$ states, not one state. The resulting s recurrent runs have state-space product at most

$$S_*^{B_{\text{res}} + s - 1} \leq S_*^{D - r} \leq S_*^d. \quad (10.5)$$

The padded profile need not have depth at most D . Only the original profile's depth is used in the guard argument. The replay theorem uses the displayed run-state product and the unchanged token/window caps.

It remains to check the removal of the source, guard, and scratch preamble. Put the actual first residual run's state immediately before the first query into $\phi(\mathbf{z}_0)$. Its subsequent underlying input is the common causal word supplied by the prefix. If $s = 1$, this is already the required one-run iid replay.

For $s \geq 2$, put $t_1 = 2q - n_{s-1}$ and $t = t_1 - w$. Then $0 < t < q$. At every query-relative position $p \geq t_1 + 1$, the local window before the second residual run begins at $p - w \geq t + 1 > 0$. It uses only first-run outputs produced on query positions. These are reconstructed from $\phi(\mathbf{z}_0)$ and the common underlying query tokens. Every earlier effect on the second run is contained in its

actual state at t_1 ; grant that state at this cut. It is a function only of $\mathbf{z}_0$ and the raw queries arrived by t_1 .

At every later cut $t_i = q + iq - n_{s-i}$, the preceding cut is more than w positions earlier. The same reconstruction applies inductively: each local window after t_i uses already reconstructed predecessor outputs, and earlier effects enter only the actual next-run state at t_i . The entire family of states is jointly causal. Thus the padded residual is a physical s -stage iid replay with full first query, original query positions, and no additional advice tape.

Theorem 9.1 gives chain success $o(1)$ on every retained chunk. Averaging over chunks and adding the discarded mass m^{-10d} preserves this bound.

10.7 Mixture and randomized models

Every avoiding profile belongs to one of the preceding cases. A profile with fewer than s local layers has reach success $o(1)$. Every other avoiding profile has chain success $o(1)$. On the unassigned branch we use only the bound one. The equal mixture consequently has success at most $1/2 + o(1)$ and error at least $1/2 - o(1)$, eventually at least $3/8$.

All table events were fixed before the competitor, its source partition, and its maps. An input-independent random seed fixes a deterministic competitor covered by the same uniform statement. Apply the bound for each seed and then average. This completes the proof of Theorem 1.1.

A Jointly causal latent-state simulation

The following elementary observation records precisely what a latent old source may do. It is useful for comparing the supplement's replay class with a representation of the actual competing network.

Lemma A.1 (One common seed preserves all later cuts). *Fix a source-independent label and a legitimate early row. Let $\mathbf{k}$ be the current live key, V any old latent data, and $\mathbf{Y}$ the future raw queries. Suppose V is conditionally independent of $\mathbf{Y}$ given the label, early row, and $\mathbf{k}$. Suppose the representative's underlying tape and transition rules are fixed by the label, early row, and arrived future, independently of V . If the remaining old dependence occurs only through frontier states $\psi_i(V, \mathbf{Y}^{1:t_i})$, their joint law can be represented by states $\tilde{\psi}_i(\mathbf{k}, U_0, \mathbf{Y}^{1:t_i})$ using one common independent seed U_0 .*

Proof. For every supported key, use one uniform seed to sample the conditional law of V ; write the resulting value as $v(\mathbf{k}, U_0)$. Define $\tilde{\psi}_i(\mathbf{k}, U_0, \mathbf{Y}^{1:t_i}) = \psi_i(v(\mathbf{k}, U_0), \mathbf{Y}^{1:t_i})$. The same sampled old data are used at all cuts, preserving their joint distribution and each information restriction. On unsupported keys choose arbitrary legal defaults. For finite data spaces this is inverse-transform sampling; an arbitrary shared seed can be reduced to the induced finite joint conditional law. $\square$

This lemma does not establish a lower bound on the fresh success of actual grants from their training success. That separate implication is Proposition 7.3 and uses the own-tested optimizing policy. A latent variable that changes the future underlying tape after the live key is made uniform is not covered by Lemma A.1.

B Interface with the shorter manuscript argument

The manuscript can use Theorem 9.1 as the technical input and retain only the two task definitions, the target implementation, and the three-case architectural reduction. The physical replay definition and the uniformity over source chunks and post-table maps must be retained in the statement or explicitly referenced. No training-tree notation is required in the manuscript’s main proof overview.

For combining the four size-two-cover components, use a common logarithmic base budget before taking the minimum of their separation factors. In particular, the other components can be constructed at the same sufficiently large b_m used here; a larger upper budget should not be introduced only after a lower bound at a smaller budget has been proved. Then choose the common separation factor no larger than each component’s factor. A constant selector/tag overhead can be accommodated by choosing a strictly smaller factor greater than one and taking m sufficiently large. The component theorem here does not by itself verify the separate wrong-order/insufficient-profile mixture or its resource normalization.

The main interfaces are Definitions 3.1 and 3.2 for physical replay; Theorem 9.1 for the uniform random chain; Lemmas 4.3 and 4.4 for first-key balance; Lemma 7.2 and Proposition 7.3 for the training implication; and equations (8.2)–(8.5) for the complete-label and crossed-continuation argument.